

Naive Bayes

Lecture 1 – Statistical Recap

Machine Learning

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1. Probability: Essential Notation

Let A and B be events.

Notation	Meaning
$P(A)$	Probability of A
$P(A^c)$	Probability of complement of A
$P(A \cap B)$	Joint probability of A and B
$P(A \cup B)$	Probability of A or B
$P(A B)$	Conditional probability of A given B
$P(A, B)$	Joint probability of A and B

Important identities

$$P(A^c) = 1 - P(A)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cap B) = P(A, B)$$

2. Conditional Probability

$$P(A | B) = \frac{P(A \cap B)}{P(B)} \quad P(B) > 0$$

Multiplication Rule

$$P(A, B) = P(A | B)P(B)$$

$$P(A, B) = P(B | A)P(A)$$

Independence

If A and B are independent:

$$P(A \cap B) = P(A)P(B)$$

and equivalently,

$$P(A | B) = P(A)$$

3. Bayes' Theorem: The Basic Idea

Suppose we want to find the probability of a class C_k after observing some information X .

Bayes' theorem is:

$$P(C_k | X) = \frac{P(X | C_k) P(C_k)}{P(X)}$$

Term	Bayesian Name	Simple Meaning
$P(C_k)$	Prior	Probability of class C_k before observing X
$P(X C_k)$	Likelihood	Probability of observing X if class is C_k
$P(X)$	Evidence	Overall probability of observing X
$P(C_k X)$	Posterior	Probability of class C_k after observing X

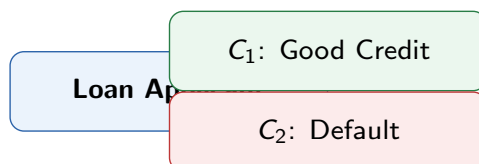
Remember

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

4. Bayes' Theorem: A Real-Life Example

Credit Risk Classification

A bank wants to classify a loan applicant into:



Suppose we observe information X about an applicant.
From historical data:

$$P(C_1) = 0.80, \quad P(C_2) = 0.20$$

and

$$P(X | C_1) = 0.10, \quad P(X | C_2) = 0.60$$

Here, X could represent a particular observed characteristic of the applicant.

4. Bayes' Theorem: Applying the Example

We want to know:

$$P(C_2 | X)$$

That is:

"Given the observed information X , what is the probability that the applicant belongs to the Default class?"

First calculate the evidence:

$$\begin{aligned} P(X) &= P(X | C_1)P(C_1) + P(X | C_2)P(C_2) \\ &= (0.10)(0.80) + (0.60)(0.20) \\ &= 0.08 + 0.12 \\ &= \boxed{0.20} \end{aligned}$$

Now apply Bayes' theorem:

$$\begin{aligned} P(C_2 | X) &= \frac{P(X | C_2)P(C_2)}{P(X)} \\ &= \frac{(0.60)(0.20)}{0.20} = \boxed{0.60} \end{aligned}$$

Decision

The larger posterior probability is $P(C_2 | X) = 0.60$. Thus, based on X , the applicant is classified into C_2 .

Similarly,

$$\begin{aligned} P(C_1 | X) &= \frac{P(X | C_1)P(C_1)}{P(X)} \\ &= \frac{(0.10)(0.80)}{0.20} = \boxed{0.40} \end{aligned}$$

Therefore:

$$\boxed{P(C_1 | X) = 0.40}$$

$$\boxed{P(C_2 | X) = 0.60}$$

4. Finding the Overall Probability $P(X)$

Suppose there are K possible classes:

$$C_1, C_2, \dots, C_K$$

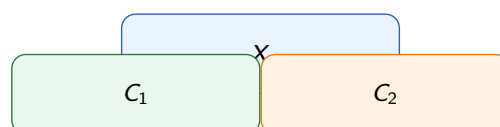
The observation X can come from any one of these classes.

Therefore, the overall probability of observing X is:

$$\boxed{P(X) = \sum_{k=1}^K P(X | C_k)P(C_k)}$$

For two classes, say C_1 and C_2 :

$$\boxed{P(X) = P(X | C_1)P(C_1) + P(X | C_2)P(C_2)}$$



5. Bayes Classifier

Suppose an observation X has to be assigned to one of the classes:

$$C_1, C_2, \dots, C_K$$

For each class, calculate:

$$P(C_k | X)$$

Then choose the class having the **largest posterior probability**.

$$\hat{C} = \arg \max_{C_k} P(C_k | X)$$

$\arg \max$ means “choose the class with the highest probability.”

Using Bayes' theorem:

$$P(C_k | X) = \frac{P(X | C_k)P(C_k)}{P(X)}$$

Since $P(X)$ is the same for every class:

$$\hat{C} = \arg \max_{C_k} P(X | C_k)P(C_k)$$

In simple words

Calculate the posterior probability for each class → choose the largest.

Naive Bayes

Lecture 2 – Introduction to Naive Bayes

Machine Learning

Pranav T V

1. From Bayes' Theorem to Classification

Suppose an observation X is to be classified into one of the classes

$$C_1, C_2, \dots, C_K.$$

Bayes' theorem gives the posterior probability of each class:

$$P(C_k | X) = \frac{P(X | C_k)P(C_k)}{P(X)}$$

For each class, we calculate:

$$P(C_1 | X), \quad P(C_2 | X), \quad \dots, \quad P(C_K | X).$$

Classification Idea

Assign X to the class having the **largest posterior probability**.

2. What is a Bayes Classifier?

A **Bayes classifier** assigns an observation X to the class with the largest posterior probability.

The classification rule is

$$\hat{C} = \arg \max_{C_k} P(C_k | X)$$

Meaning of arg max

Choose the class C_k that gives the largest value of $P(C_k | X)$.

Using Bayes' theorem,

$$P(C_k | X) = \frac{P(X | C_k)P(C_k)}{P(X)}.$$

Since $P(X)$ is common to all classes,

$$\hat{C} = \arg \max_{C_k} P(X | C_k)P(C_k)$$

3. Classification with Multiple Features

In practice, an observation usually contains several features.

Let

$$X = (X_1, X_2, \dots, X_p)$$

where $X_1, \dots, X_p$ are the observed features.

Bayes' theorem becomes

$$P(C_k | X_1, \dots, X_p) = \frac{P(X_1, \dots, X_p | C_k)P(C_k)}{P(X_1, \dots, X_p)}.$$

Therefore, the Bayes classification rule is

$$\hat{C} = \arg \max_{C_k} P(X_1, \dots, X_p | C_k)P(C_k)$$

Main Difficulty

The term

$$P(X_1, \dots, X_p | C_k)$$

is a joint conditional probability involving all features.

Naive Bayes

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4. What is Naive Bayes?

Naive Bayes is a classification method based on Bayes' theorem.

It makes one important simplifying assumption:

Features are conditionally independent given the class.

For

$$X = (X_1, X_2, \dots, X_p),$$

the assumption gives

$$P(X_1, \dots, X_p | C_k) = \prod_{j=1}^p P(X_j | C_k)$$

That is,

$$P(X_1, \dots, X_p | C_k) = P(X_1 | C_k)P(X_2 | C_k) \cdots P(X_p | C_k).$$

Key Idea

Naive Bayes replaces a complicated joint probability with a product of simpler probabilities.

Naive Bayes

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5. Why is it Called “Naive”?

The method is called **Naive Bayes** because of its simplifying assumption about the features.

Suppose

$$X_1 = \text{Income}, \quad X_2 = \text{Debt}, \quad X_3 = \text{Employment Status}.$$

Let C denote the credit-risk class.

Naive Bayes assumes that the features are conditionally independent given the class:

$$X_1 \perp X_2 \perp X_3 \mid C$$

This means

$$P(X_1, X_2, X_3 \mid C) = P(X_1 \mid C)P(X_2 \mid C)P(X_3 \mid C)$$

Why “Naive”?

The assumption may not always be realistic, but it makes the probability calculation much simpler.

6. Conditional Independence

Two features X_1 and X_2 are conditionally independent given a class C if

$$P(X_1, X_2 \mid C) = P(X_1 \mid C)P(X_2 \mid C)$$

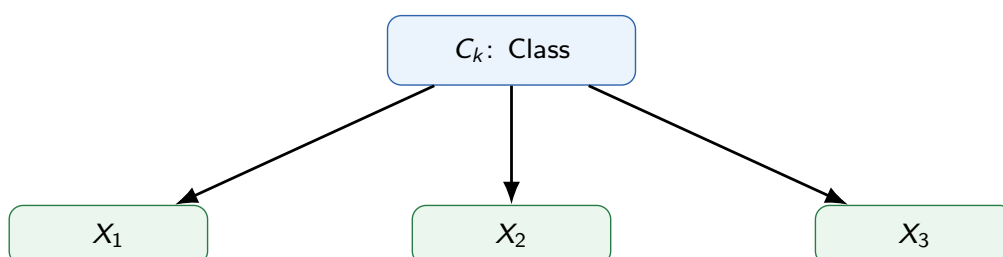
The important phrase is:

“given the class”

For several features,

$$P(X_1, \dots, X_p \mid C_k) = \prod_{j=1}^p P(X_j \mid C_k)$$

A simple representation:



7. Naive Bayes Classification Rule

The Bayes classifier is

$$\hat{C} = \arg \max_{C_k} P(X_1, \dots, X_p | C_k) P(C_k).$$

Under the Naive Bayes assumption,

$$P(X_1, \dots, X_p | C_k) = \prod_{j=1}^p P(X_j | C_k).$$

Therefore,

$$\hat{C} = \arg \max_{C_k} \left[P(C_k) \prod_{j=1}^p P(X_j | C_k) \right]$$

Equivalently,

$$\hat{C} = \arg \max_{C_k} P(C_k) P(X_1 | C_k) \cdots P(X_p | C_k)$$

Naive Bayes Rule

Prior probability $\times$ product of feature likelihoods

8. Naive Bayes: Classification Process

For a new observation

$$X = (X_1, X_2, \dots, X_p),$$

Naive Bayes follows the following steps.



The final decision is

Choose the class with the largest class score.

9. Basic Terminology

Term	Notation	Meaning
Class	C_k	Category that we want to predict
Feature	X_j	Observed characteristic used for classification
Prior	$P(C_k)$	Probability of the class before observing the features
Likelihood	$P(X_j \mid C_k)$	Probability of observing a feature value given the class
Posterior	$P(C_k \mid X)$	Probability of the class after observing the features

For Naive Bayes,

$$P(C_k \mid X_1, \dots, X_p) \propto P(C_k) \prod_{j=1}^p P(X_j \mid C_k)$$

10. Simple Two-Class Example

Consider a bank classifying a loan applicant into one of two classes:

Class	Meaning
C_1	Good Credit
C_2	Default

Suppose two features are observed:

$$X_1 = \text{High Income}, \quad X_2 = \text{Low Debt}.$$

Thus,

$$X = (X_1, X_2).$$

We want to decide whether the applicant belongs to

$$C_1 \quad \text{or} \quad C_2$$

11. Example: Given Probabilities

Suppose historical data gives the class probabilities:

$$P(C_1) = 0.80, \quad P(C_2) = 0.20.$$

The feature likelihoods are:

$$P(X_1 | C_1) = 0.70, \quad P(X_2 | C_1) = 0.80,$$

$$P(X_1 | C_2) = 0.30, \quad P(X_2 | C_2) = 0.40.$$

Under the Naive Bayes assumption,

$$P(X_1, X_2 | C_k) = P(X_1 | C_k)P(X_2 | C_k).$$

Therefore, the class score for C_1 is

$$\begin{aligned} P(C_1)P(X_1 | C_1)P(X_2 | C_1) &= (0.80)(0.70)(0.80) \\ &= \boxed{0.448}. \end{aligned}$$

12. Example: Comparing the Class Scores

For class C_2 ,

$$\begin{aligned} P(C_2)P(X_1 | C_2)P(X_2 | C_2) &= (0.20)(0.30)(0.40) \\ &= \boxed{0.024}. \end{aligned}$$

Therefore, the two class scores are:

$$\boxed{C_1 : 0.448}$$

$$\boxed{C_2 : 0.024}$$

Since

$$0.448 > 0.024,$$

the Naive Bayes rule gives

$$\boxed{\hat{C} = C_1}$$

Thus, the applicant is classified as

Good Credit.

Key Point

Only the relative size of the class scores is needed to determine the predicted class.

13. Why Does Naive Bayes Work?

The main computational advantage comes from the independence assumption.

Without the Naive Bayes assumption:

$$P(X_1, X_2, \dots, X_p \mid C_k)$$

must be estimated as a joint probability.

With the assumption:

$$P(X_1, \dots, X_p \mid C_k) = \prod_{j=1}^p P(X_j \mid C_k)$$

Thus, instead of estimating one complicated joint distribution, we estimate several simpler conditional distributions:

$$P(X_1 \mid C_k), \quad P(X_2 \mid C_k), \quad \dots, \quad P(X_p \mid C_k).$$

Main Advantage

The model becomes much simpler to estimate and compute.

14. Bayes Classifier vs. Naive Bayes

	Bayes Classifier	Naive Bayes
Basic idea	Uses Bayes' theorem for classification	Uses Bayes' theorem with a conditional independence assumption
Features	Uses the joint distribution of the features	Treats features as conditionally independent given the class
Conditional probability	$P(X_1, \dots, X_p \mid C_k)$	$\prod_{j=1}^p P(X_j \mid C_k)$
Main advantage	Can represent dependence between features	Simpler probability estimation

Main Difference

Naive Bayes is a simplified Bayes classifier.

15. Summary: Introduction to Naive Bayes

1 Bayes classifier:

$$\hat{C} = \arg \max_{C_k} P(C_k | X).$$

2 Multiple features:

$$X = (X_1, \dots, X_p).$$

3 Naive Bayes assumption:

$$P(X_1, \dots, X_p | C_k) = \prod_{j=1}^p P(X_j | C_k).$$

4 Naive Bayes classification rule:

$$\hat{C} = \arg \max_{C_k} \left[P(C_k) \prod_{j=1}^p P(X_j | C_k) \right]$$

Core Idea

Bayes' theorem + Conditional independence = Naive Bayes

10. Naive Bayes Example: Playing Golf

Consider the following dataset describing weather conditions and whether it was suitable to play golf.

Weather	Temperature	Play?
Sunny	Hot	No
Sunny	Hot	No
Overcast	Hot	Yes
Rainy	Mild	Yes
Rainy	Cool	Yes
Rainy	Cool	No
Overcast	Cool	Yes
Sunny	Mild	No
Sunny	Cool	Yes
Rainy	Mild	Yes
Sunny	Mild	Yes
Overcast	Mild	Yes
Overcast	Hot	Yes
Rainy	Mild	No

There are two possible classes: $C_1 = \text{Play} = \text{Yes}$, $C_2 = \text{Play} = \text{No}$.

Goal

Predict whether we can play golf when

11. The New Observation

We are given a new observation:

Weather = Sunny

Temperature = Mild

We need to compare the probabilities:

$$P(\text{Play} = \text{Yes} \mid \text{Sunny, Mild})$$

and

$$P(\text{Play} = \text{No} \mid \text{Sunny, Mild}).$$

Using the Naive Bayes assumption,

$$P(X_1, X_2 \mid C_k) = P(X_1 \mid C_k)P(X_2 \mid C_k)$$

Therefore, we calculate the two class scores separately.

Decision Rule

The class with the larger score will be the prediction.

Step 1: Calculate Prior Probabilities

The first step is to calculate the probability of each class.

From the dataset:

Total number of observations = 14

There are:

9 observations with Play = Yes

and

5 observations with Play = No.

Therefore,

$$P(\text{Play} = \text{Yes}) = \frac{9}{14} \quad \text{and} \quad P(\text{Play} = \text{No}) = \frac{5}{14}$$

Prior Probability

The prior probability tells us how common each class is before considering the new observation.

Step 2: Calculate Likelihood Probabilities

Now calculate the likelihood of each observed feature within each class.

For Weather = Sunny:

$$P(\text{Sunny} \mid \text{Play} = \text{Yes}) = \frac{2}{9}$$

because 2 of the 9 “Yes” observations are Sunny.

$$P(\text{Sunny} \mid \text{Play} = \text{No}) = \frac{3}{5}$$

because 3 of the 5 “No” observations are Sunny.

For Temperature = Mild:

$$P(\text{Mild} \mid \text{Play} = \text{Yes}) = \frac{4}{9}$$

and

$$P(\text{Mild} \mid \text{Play} = \text{No}) = \frac{2}{5}.$$

Likelihoods

Likelihoods measure how frequently the observed feature values occur within each class.

Step 3: Apply Naive Bayes – Play = Yes

For the class

$$C_1 = \text{Play} = \text{Yes},$$

the Naive Bayes class score is

$$P(\text{Sunny} \mid \text{Play} = \text{Yes}) P(\text{Mild} \mid \text{Play} = \text{Yes}) P(\text{Play} = \text{Yes}).$$

Substituting the values:

$$\begin{aligned} &P(\text{Sunny} \mid \text{Play} = \text{Yes}) P(\text{Mild} \mid \text{Play} = \text{Yes}) P(\text{Play} = \text{Yes}) \\ &= \frac{2}{9} \times \frac{4}{9} \times \frac{9}{14} \end{aligned}$$

Therefore,

$$\frac{2}{9} \times \frac{4}{9} \times \frac{9}{14} = \frac{8}{126} = 0.0635$$

Hence,

$$\text{Score}(\text{Play} = \text{Yes}) = 0.0635$$

Step 3: Apply Naive Bayes – Play = No

For the class

$$C_2 = \text{Play} = \text{No},$$

the Naive Bayes class score is

$$P(\text{Sunny} \mid \text{Play} = \text{No}) P(\text{Mild} \mid \text{Play} = \text{No}) P(\text{Play} = \text{No}).$$

Substituting the values:

$$\begin{aligned} &P(\text{Sunny} \mid \text{Play} = \text{No}) P(\text{Mild} \mid \text{Play} = \text{No}) P(\text{Play} = \text{No}) \\ &= \frac{3}{5} \times \frac{2}{5} \times \frac{5}{14} \end{aligned}$$

Therefore,

$$\frac{3}{5} \times \frac{2}{5} \times \frac{5}{14} = \frac{30}{350} = 0.0857$$

Hence,

$$\text{Score}(\text{Play} = \text{No}) = 0.0857$$

Step 4: Make the Prediction

We have obtained the two class scores:

$$\text{Play} = \text{Yes} : 0.0635$$

$$\text{Play} = \text{No} : 0.0857$$

Compare the two values:

$$0.0857 > 0.0635$$

Therefore,

$$\hat{C} = \text{Play} = \text{No}$$

Prediction: Play = No

Naive Bayes Example: Summary of Steps

For the new observation

$$X = (\text{Sunny}, \text{Mild})$$

we followed four steps.

1 Calculate Prior Probabilities

$$P(\text{Yes}) = \frac{9}{14}, \quad P(\text{No}) = \frac{5}{14}$$

2 Calculate Likelihoods

$$P(\text{Sunny} \mid \text{Yes}) = \frac{2}{9}, \quad P(\text{Mild} \mid \text{Yes}) = \frac{4}{9}$$

$$P(\text{Sunny} \mid \text{No}) = \frac{3}{5}, \quad P(\text{Mild} \mid \text{No}) = \frac{2}{5}$$

3 Calculate Class Scores

$$\text{Yes} : 0.0635, \quad \text{No} : 0.0857$$

4 Choose the Larger Score

$$\hat{C} = \text{Play} = \text{No}$$

What Did the Naive Assumption Do?

For the two observed features,

$$X_1 = \text{Sunny}, \quad X_2 = \text{Mild},$$

the Naive Bayes assumption allows us to write

$$P(\text{Sunny}, \text{Mild} \mid C_k) = P(\text{Sunny} \mid C_k)P(\text{Mild} \mid C_k)$$

Therefore, instead of directly estimating

$$P(\text{Sunny}, \text{Mild} \mid C_k),$$

we estimate two simpler probabilities:

$$P(\text{Sunny} \mid C_k)$$

and

$$P(\text{Mild} \mid C_k).$$

Core Idea

The Naive Bayes assumption allows the joint likelihood to be written as a product of individual feature likelihoods.

Naive Bayes

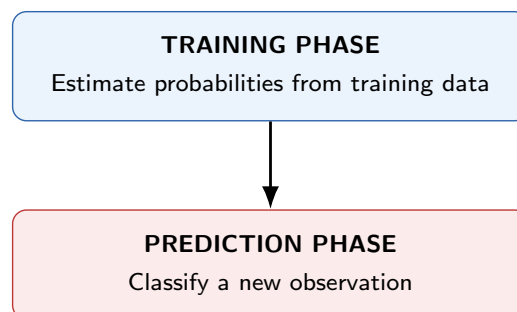
Lecture 3 – Naive Bayes Algorithm – Working and Classification

Machine Learning

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1. Naive Bayes Algorithm: Overview

Naive Bayes classification consists of two main stages:



Training Phase

Calculate the class prior probabilities and feature likelihoods from the training data.

Prediction Phase

Use these probabilities to calculate a score for each class and select the class with the largest score.

2. Training Phase

Suppose the training data contains:

$$X = (X_1, X_2, \dots, X_p)$$

as features and

$$C_1, C_2, \dots, C_K$$

as the possible classes.

The training phase calculates two quantities.

❶ Prior probability of each class

$$P(C_k)$$

This tells us how common class C_k is in the training data.

❷ Likelihood of each feature given the class

$$P(X_j | C_k)$$

This tells us how likely feature X_j is when the observation belongs to class C_k .

Training Output

The model stores the estimated values of $P(C_k)$ and $P(X_j | C_k)$.

3. Prediction Phase: New Observation

After training, suppose a new observation is given:

$$X = (X_1, X_2, \dots, X_p)$$

The objective is to determine which class C_k the new observation belongs to.

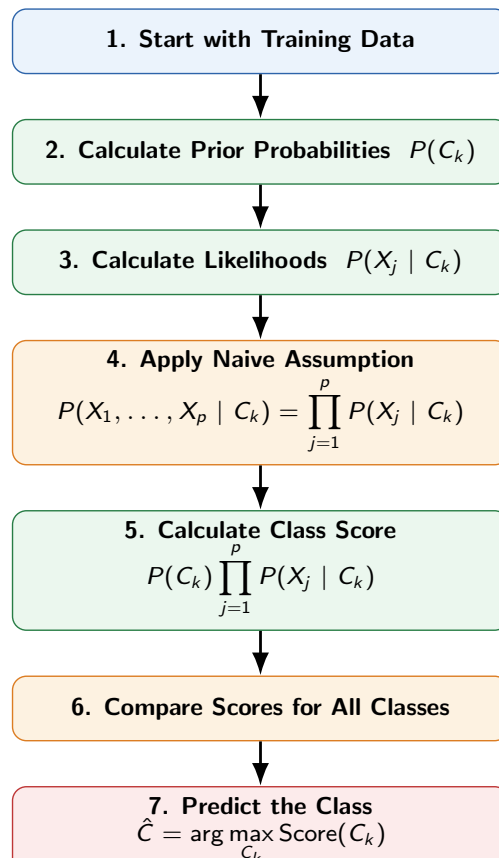
For every class C_k , calculate:

$$P(C_k) \prod_{j=1}^p P(X_j | C_k).$$

This gives the **Naive Bayes class score**:

$$\text{Score}(C_k) = P(C_k) \prod_{j=1}^p P(X_j | C_k)$$

Finally, compare the scores for all classes.



5. Naive Bayes Algorithm: Flowchart

In Simple Words

Observe features → calculate probabilities → calculate class scores → choose the largest score

7. Naive Bayes in a Real-Life Example

Consider **email spam detection**.

A new email is observed with features such as:

X_1 = contains “free”,

X_2 = contains several links,

X_3 = unknown sender.

The possible classes are:

C_1 = Spam, C_2 = Not Spam.

Naive Bayes calculates:

$$\text{Score}(C_1) = P(C_1)P(X_1 | C_1)P(X_2 | C_1)P(X_3 | C_1)$$

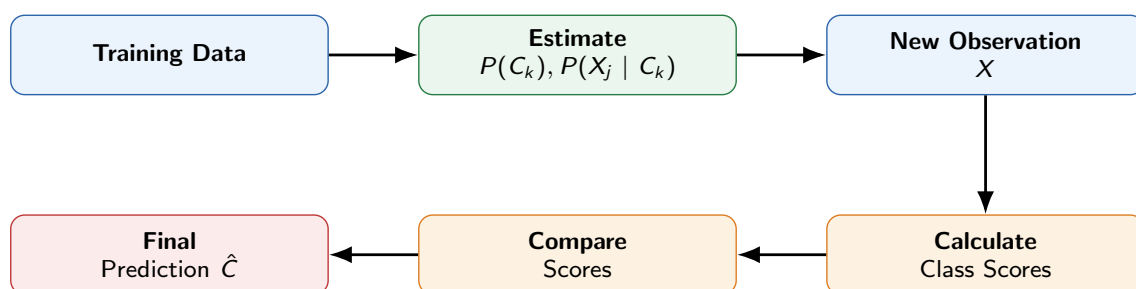
and

$$\text{Score}(C_2) = P(C_2)P(X_1 | C_2)P(X_2 | C_2)P(X_3 | C_2).$$

Decision

The email is assigned to the class with the larger score.

8. Naive Bayes: Complete Workflow



Core Idea

Learn probabilities from past data, then use them to classify a new observation.